project

Sunday, May 10, 2026 6:03 AM

Assumptions: $\dot{m}$, k, c_p, $T_{h,i}$, $T_{h,e}$, μ

Guesses: T_{ho} = ☐, T_{co} = ☐

Unknowns: q, L, ε ↑ iterate! (UA)

① (from $E_{balance}$) $\begin{cases} q = C_c(T_{co} - T_{ci}) \\ q = C_h(T_{hi} - T_{ho}) \end{cases}$ ★

② for LMTD $\begin{cases} \Delta T_1 = T_{hi} - T_{co} \\ \Delta T_2 = T_{ho} - T_{ci} \end{cases}$ $\Delta T_{lm} = \dfrac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$

∴ $UA = \dfrac{q}{\Delta T_{lm}}$ ("overall heat transfer conductance")

by NTU: ideal transfer rate

③ $q = \varepsilon C_{min}(T_{h,i} - T_{ci})$

$NTU = \frac{UA}{C_{min}}$ = # vs counter

$Re = \dfrac{4\dot{m}}{\pi D \mu}$ $Nu = \dfrac{hD}{k}$

by E balance: $q_{h,lost} = q_{regained}$

① $\begin{cases} q = C_c(T_{co} - T_{ci}) \\ q = C_h(T_{hi} - T_{ho}) \end{cases}$ →

by LMTD: $q = UA\Delta T_{lm}$ ②

③ find U? for both hot/cold sides Re → Nu → h → U

$Re = \dfrac{\rho V D_h}{\mu}$ → $Nu = \begin{cases} 0.023\, Re^{0.8} Pr^n & \text{(turbulent)} \\ 3.66 & \text{(laminar)} \end{cases}$

annulus: $D_h = D_{out} - D_{in}$

$U = \dfrac{D_o}{D_i}\dfrac{1}{h_i} + \dfrac{D_o \ln\left(\frac{D_o}{D_i}\right)}{2k_t} + \dfrac{1}{h_o}$ too many variables

$A = \dfrac{\pi}{4}(D_{outside}^2 - D_{inner}^2)$

④ set $q_{NTU} = q_{LMTD}$ → find at end $A_{req} = \pi D_o L$ → $L = \dfrac{A_{req}}{\pi D_o}$

★ too many variables assume T_{out} at first, AND across different efficiencies

↳ 3.5 heat capacity section

$q_{max} = C_{min}(T_{hi} - T_{ci})$

∴ $q = \varepsilon q_{max} = \varepsilon C_{min}(T_{hi} - T_{ci})$

EQUATION SUMMARY

E_{bal} $\begin{cases} q = C_c(T_{co} - T_{ci}) \\ q = C_h(T_{hi} - T_{ho}) \end{cases}$ MAIN LOOP

NTU $q = \varepsilon C_{min}(T_{h,i} - T_{ci})$ → $\varepsilon = \dfrac{h}{C_{min}(T_{hi} - T_{ci})}$ ∴ $NTU = \dfrac{\ln\left(\frac{1 - \varepsilon C_r}{1 - \varepsilon}\right)}{1 - C_r}$

LMTD $q = UA\Delta T_{lm}$ ← lump UA's (for now) ∴ $UA_{NTU} = NTU\, C_{min}$

★ $UA_{NTU} = UA_{LMTD}$

$\begin{cases} U = \dfrac{D_o}{D_i}\dfrac{1}{h_i} + \dfrac{D_o \ln\left(\frac{D_o}{D_i}\right)}{2k_t} + \dfrac{1}{h_o} \\ A = \dfrac{\pi}{4}(D_{outside}^2 - D_{inner}^2) \end{cases}$

$Re = \dfrac{\rho V D_h}{\mu}$ → $Nu = \begin{cases} 0.023\, Re^{0.8} Pr^n & \text{(turbulent)} \\ 3.66 & \text{(laminar)} \end{cases}$

$h = \dfrac{Nuk}{D}$

DESIGN SWEEP

set D_i show U

let t = thickness (assume?)

$D_o = D_i + 2t$